

AI-Powered Credit Risk Intelligence Platform

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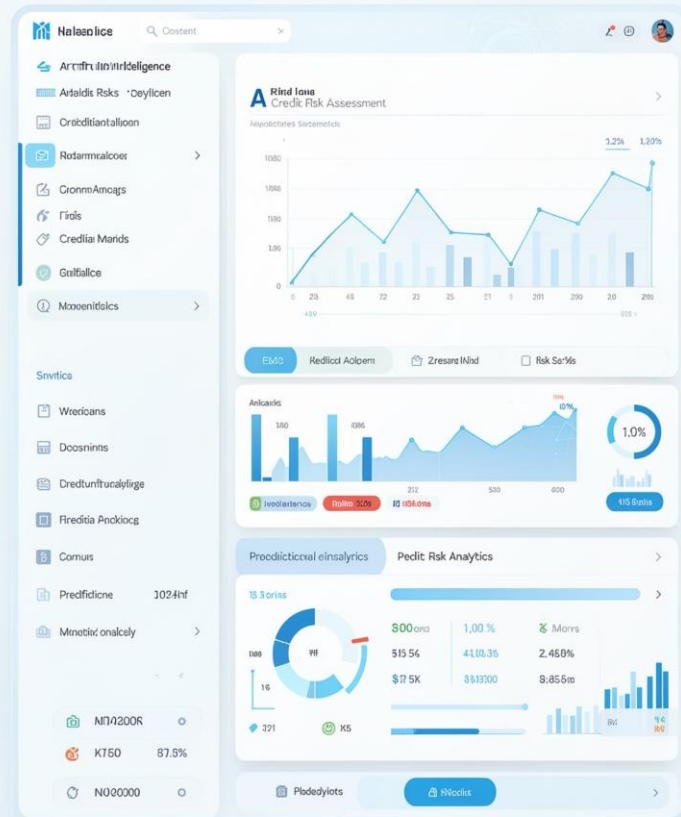
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Introduction

Credit risk assessment is critical in the financial industry. Traditional methods rely on manual analysis and predefined rules, which may not capture complex patterns. This platform combines machine learning, explainable AI, business rules, and conversational analytics to support credit decision-making.



Problem Statement

Accurately identifying high-risk loan applicants while maintaining transparency in decision-making. Existing methods may be time-consuming, less scalable, and difficult to interpret.

Objectives

- Predicting applicant default risk using machine learning
- Providing explainable predictions
- Supporting natural language querying of credit data
- Generating business rules for risk assessment
- Visualizing important credit risk insights



Dataset Overview

The project utilizes the Home Credit Default Risk dataset containing information about loan applicants. Target variable: TARGET = 0 (Non-Default), TARGET = 1 (Default).

Metric	Value
Total Records	307,511
Initial Features	122
Final Features	80
Default Rate	8.07%

Exploratory Data Analysis

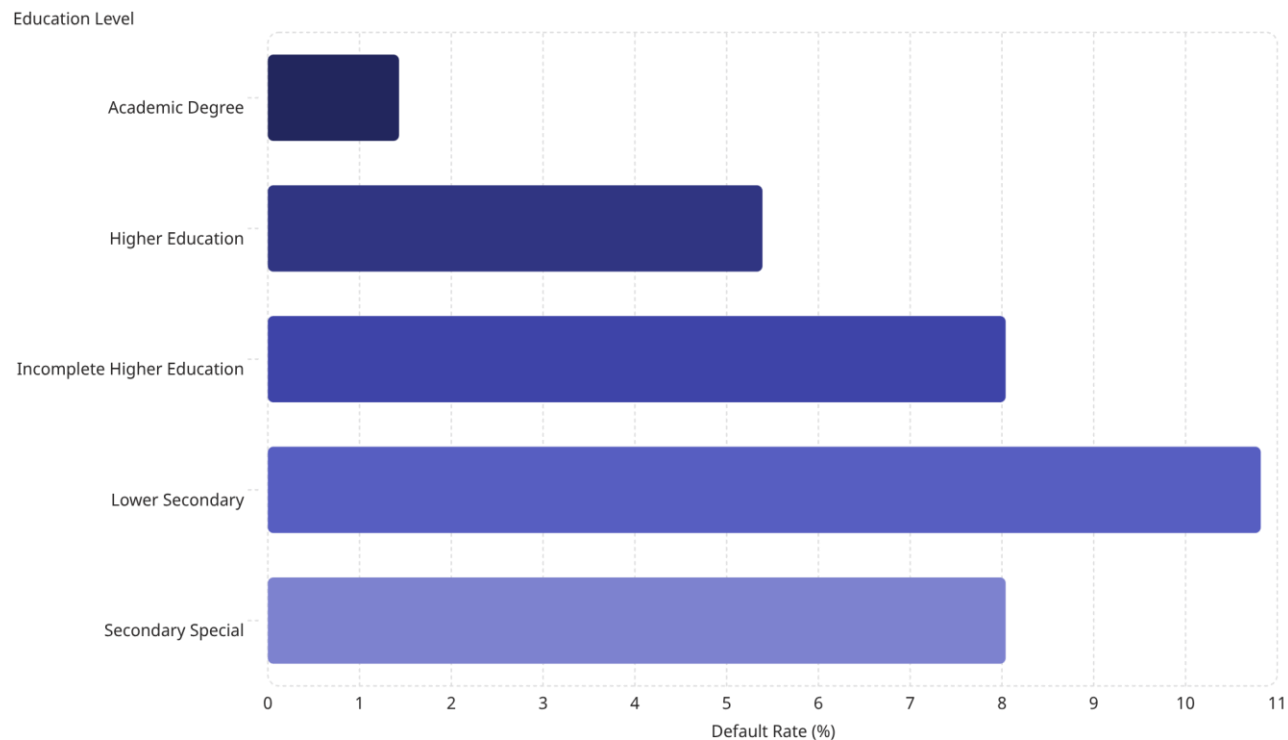
Key findings from the Home Credit Default Risk dataset:

- Default rate is approximately 8.07%
- Male applicants exhibit higher default rates than female applicants
- Applicants with lower secondary education have the highest default rate
- Academic degree holders have the lowest default rate
- Credit amount and external risk scores strongly influence default probability



This chart illustrates the distribution of loan applicants based on their default status. A significant majority of applicants (91.93%) are non-defaulting, while a smaller portion (8.07%) represent default cases, highlighting the class imbalance in the dataset.

Education-wise Default Analysis



Education level is a significant factor in credit risk assessment.

Data Preprocessing

01

Missing value treatment

02

Feature selection

03

Numerical imputation

04

Categorical encoding

05

Train-test split

06

Pipeline construction

The dataset was split into training and testing sets using an 80:20 ratio.



Machine Learning Model

Multiple machine learning approaches were considered, and XGBoost was selected due to its superior predictive performance.

Model Training

- Scikit-Learn Pipeline
- Column Transformer
- XGBoost Classifier



Model Performance

Model performance was evaluated using the ROC-AUC metric. The obtained ROC-AUC score of 0.753 demonstrates the model's ability to distinguish between default and non-default applicants effectively.

Metric	Value
ROC-AUC Score	0.753

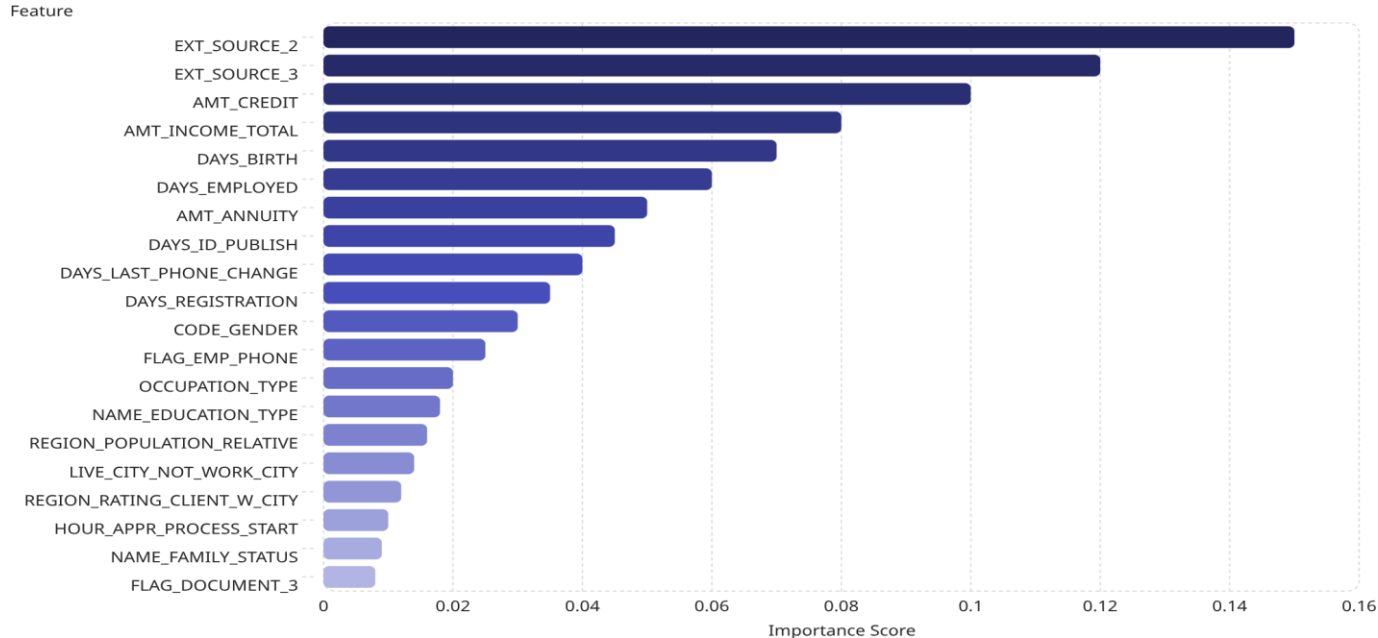
0.753

ROC-AUC Score

Distinguishes between default and non-default applicants effectively.

Explainable AI

To improve transparency and trust in model predictions, explainability techniques were incorporated. Feature importance analysis was performed to identify variables that most strongly influence loan default predictions.



This bar chart illustrates the top 20 most important features identified by the model that influence loan default predictions. External risk scores (EXT_SOURCE_2, EXT_SOURCE_3), credit amount (AMT_CREDIT), total income (AMT_INCOME_TOTAL), and applicant's age (DAYS_BIRTH) are among the strongest indicators.

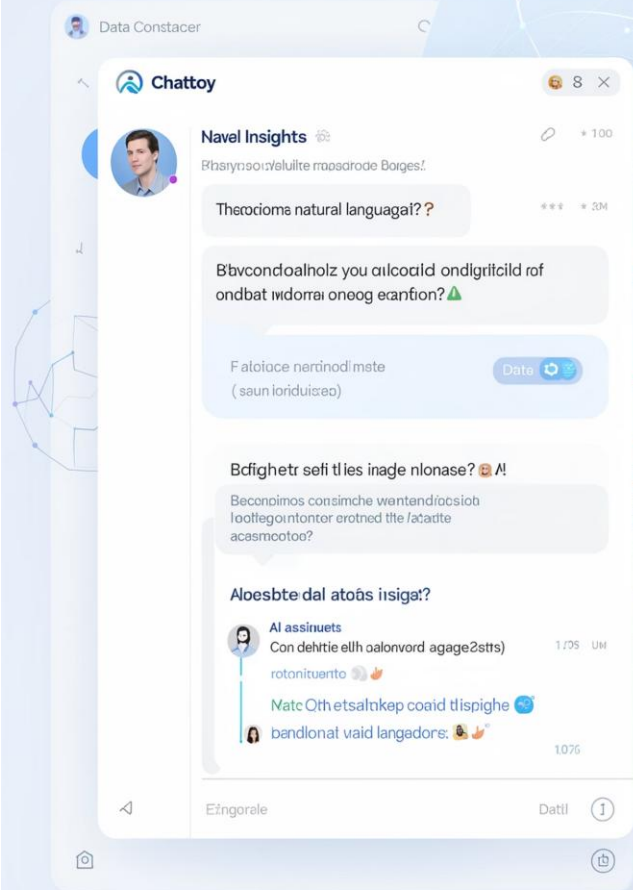
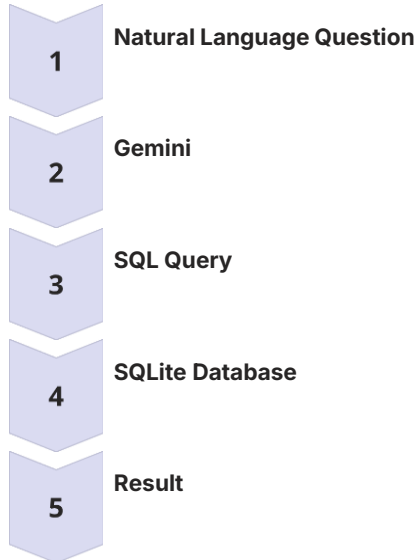
Talk-to-Data Chatbot

A conversational analytics module was developed using Google Gemini and SQLite. The chatbot converts natural language questions into SQL queries and retrieves answers from the credit risk database.

Example Questions

- How many customers defaulted?
- What is the average applicant income?
- Which education group has the highest default rate?
- Which gender has the highest default rate?

Chatbot Workflow



Streamlit Dashboard

A web-based dashboard was developed using Streamlit to provide an interactive user experience.



Home Page



EDA Dashboard



Credit Risk Prediction



Explainability



Business Rules



Talk-to-Data Chatbot



Conclusion

The AI-Powered Credit Risk Intelligence Platform successfully integrates machine learning, explainable AI, business intelligence, and conversational analytics into a unified decision-support system. The developed solution achieved a ROC-AUC score of 0.753 using XGBoost and provides transparent, explainable, and user-friendly credit risk assessment capabilities. The platform demonstrates how modern AI technologies can enhance lending decisions, improve operational efficiency, and support risk management in financial institutions.

